CS-229: K-means clustering:-

- The core idea is ~~so~~ short: - K-means groups ~~unlabe~~ unlabeled data by repeatedly doing two things - assign each point to its nearest centroid, then move each centroid to the mean of its assigned points.

- The two steps, concretely:
  Say u have 6 points & $K = 2$ centroids at $\mu_1 = (2,$
  & $\mu_2 = (8, 8)$.

Step-I: - Assign $\left(c^{(i)} = \arg\min \|x^{(i)} - \mu_j\|^2 \right)$;
point $(3,3) \rightarrow$ closer to $\mu_1 \rightarrow$ gets label 1. Point $(7,9) \rightarrow$ closer to μ_2 $\rightarrow$ gets label 2. Just Euclidean dist. Pick closest centroid.

St-2 : -

move (μ_j = mean of all points assigned to cluster j): If cluster 1 has points (1,1), (2,3), (3,2) → new μ_1 = (2,2). Centroid slides to the group's center of mass.

Repeat until assignments stop changing = convergence.

why does it converge?

⇒ The distortion function ~~$J = \sum \|x^{(i)}$~~

$J = \sum \|x^{(i)} - \mu_{c^{(i)}}\|^2$ (sum of squared distances to assigned centroid can only dec(↓) or stay the same at each step - never inc(↑). Assigning each point to the nearest centroid can only lower J. moving each centroid to the mean of its cluster can only lower J. so J monotonically decreases → must converge.

But - J is non-convex. It can get stuck in a bad local minimum depending on initial centroids. that's why u run it multiple times with diff. random starts & pick the run with lowest J.

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Mixture of Gaussians — Big picture

(by seeing these points, you don't know which Gaussian each ~~each~~ come from — $z^{(i)}$ is hidden (latent))

↓

G_1: μ_1, Σ_1, ϕ_1

G_2: μ_2, Σ_2, ϕ_2

G_3: μ_3, Σ_3, ϕ_3

E-step: For each point i & Gaussian j, compute soft probability $w_j^{(i)} = p(z^{(i)} = j \mid x^{(i)})$

→

M-step: use $w_j^{(i)}$ as soft counts to re-estimate ϕ_j, μ_j, Σ_j for each Gaussian

Like k-mean but soft assignments (probabilities) instead of "hard" cluster labels repeat until convergence

Core Idea simply :- You have data points. You suspect they come from K diff Gaussian distributions, but U don't know which point came from which Gaussian → these hidden cluster labels are $z^{(i)}$ — the latent variables.

The parameters u want to find: ϕ (mixing weights — how common is each Gaussian), μ (means), Σ (covariances).

If U knew $z^{(i)}$, the problem is trivial — just compute the mean & covariance of each cluster separately. That's B's says formulas on p-2 with the indicator functions $1\{z^{(i)} = j\}$.

- You don't know $z^{(i)}$, so EM does this:
  - E-step: Guess the hidden labels softly. For every point i & every cluster j, compute $w^{(i)}_j = P(z^{(i)} = j \mid x^{(i)})$ using Bayes rule with the current parameters. This is a no b/w 0 & 1 – "how much does point i belong to cluster j?"

  - M-step: pretend those soft guesses are real, & re-estimate the parameters. Instead of hard counts $1\{z^{(i)} = j\}$, you use soft weights $w^{(i)}_j$. The formulas are identical in structure.

  - Repeat untill parameters stop changing.

- Key Intuition: ~~E-step~~ E & M are mutually bootstrapping. Better parameters → better soft assignments → better ~~perfo.~~ parameters → ...